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SUPERSEDING
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FEDERAL SPECIFICATION

MATTRESSES, BED, INNERSPRING

This specification is approved by the Assistant Administrator, Office of Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the construction and materials for innerspring bed mattresses.

1.2 Classification.

1.2.1 Types. The mattresses shall be of the following types, classes, sizes, and degree of firmness as specified (see 6.2).

Types:

Type I - (Deleted)
Type II - All-wire spring units, open end offset coil construction
Type III - All-wire spring units, knotted coil construction
Type IV - (Deleted)

Classes:

Class 1 - Without handles
Class 2 - With handles

Sizes. Sizes of mattresses shall be as shown in tables III and IV as specified (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7210

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Firmness:

Regular
Firm
Extra firm

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications

V-T-276	-	Thread, Cotton
V-T-285	-	Thread, Polyester
V-T-295	-	Thread, Nylon
CCC-C-430	-	Cloth, Sheeting, Cotton, and Polyester and Cotton
CCC-C-432	-	Cloth, Sheeting, Cotton, (Unbleached, Bleached, and Dyed)
CCC-C-436	-	Cloth, Ticking Twill, Cotton
DDD-L-20	-	Label: for Clothing, Equipage, and Tentage, (General Use)

Federal Standards

FED-STD-123	-	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-191	-	Textile Test Methods
FED-STD-751	-	Stitches, Seams, and Stitchings

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bi-monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Single copies of this specification and other Federal Specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.)

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications

MIL-B-41826	-	Batting, Synthetic Fibers: Polyester, (Quilted and Unquilted)
MIL-M-3996	-	Mattresses (Except Pneumatic Type), Packaging and Packing Of

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Military Standards

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the contracting activity or as directed by the contracting activity.)

Federal Regulations

Consumer Product Safety Commission

FF 4-72 - Standard for the Flammability of Mattresses and Mattress Pads (16 CFR Part 1632)

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal Agency responsible for issuance thereof.)

U.S. Department of Agriculture

C&MS-SRA-180 - Cotton and Cotton Linters Standards

(Application for copies should be addressed to the Cotton Division, Agricultural Marketing Service, Department of Agriculture, Washington, DC 20250.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Flammability. All mattresses shall meet the provisions of the Consumer Product Safety Commission Standard for the Flammability of Mattresses and Mattress Pads FF-4-72 (16 CFR Part 1632) (see 4.6).

3.3 Materials and components. Recycled components may be used for nontextile components (see 6.4).

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3.3.1 Cotton. The cotton shall be new (not previously used), minimum 7/8-inch staple cotton and not lower in grade than the lowest grade in the white, spotted, or gray grades, of the United States Department of Agriculture Publication No. C&MS-SRA-180. The use of reginned cotton will be permitted provided the required cotton is equivalent in grade to the standards listed above. Tinged or stained cotton shall not be permitted. Testing shall be as specified in 4.4.1.2.

3.3.2 Linters. The linters for the mattresses shall be new (not previously used) conforming to grade number 3 and staple 3 of USDA Publication No. C&MS-SRA-180. Testing shall be as specified in 4.4.1.2.

3.3.3 Polyester fiber.

3.3.3.1 Polyester fiber for grade B felt. The fiber to be blended with the linters specified in 3.3.2 for grade B felt (see 3.3.4.2) shall consist of a minimum of 20 percent mixture of crimped, (6 to 8 crimps per inch) cut staple virgin polyester fiber, of 5.0 or 6.0 denier, 2 ± 1/2 inch in length and the balance of the blend may be made from polyester fiber (95 percent pure polyester fiber) (see 4.4.1.2).

3.3.3.2 Polyester fiber for grade C felt. The fiber to be used for grade C felt (see 3.3.4.3) shall consist of 100 percent crimped, cut staple, new (not used) polyester fiber of 3.5 to 4.5 denier, 2 ± 1/2 inches in length and having a minimum melting point of 470°F (see 4.4.1.2).

3.3.4 Felt (see 3.5.6 and 3.5.7). The felt shall be clean and shall be so processed as to produce unbroken laminated sheets. The felt may be of any of the grades specified below (see 3.3.4.1, 3.3.4.2 and 3.3.4.3).

3.3.4.1 Grade A. Grade A felt shall be made of a blend of 50 percent cotton conforming to 3.3.1 and 50 percent linters conforming to 3.3.2.

3.3.4.2 Grade B. Grade B felt shall be made of a blend of 65 percent polyester fiber conforming to 3.3.2.1 and 35 percent linters conforming to 3.3.2 (see 4.4.1.2).

3.3.4.3 Grade C. Grade C felt shall be made of 100 percent polyester fiber conforming to 3.3.3.2 garnetted to form even unbroken sheets of unbonded batts (see 4.4.1.2).

3.3.5 Barrier materials.

3.3.5.1 Polyester batting. Polyester batting conforming to type III of MIL-B-41826 shall be used as a barrier, except that the requirements for thickness after laundering, compressional recovery after laundering, and launderability need not apply.

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3.3.5.2 Commercial bonded polyester batting. The use of a commercial bonded polyester batting for use as a barrier material is acceptable provided that the batting is bonded with an approved durable bonding agent (see 6.6) uniformly applied. The maximum concentration of the applied bonding agent shall not exceed 34.0 percent on the weight of the finished bonded batting. The bonding agent shall be applied to both sides of the batting filler.

3.3.5.3 Approval and qualification of other barrier materials (see 3.5.7). Other commercial type mattress barrier materials or smoldering resistant treatments may also be used as barrier materials. Approval of these materials and or treatments must be obtained prior to use (see 6.5). The use of an impermeable material as a barrier material is prohibited (see 4.4.1).

3.3.6 Ticking. The ticking shall conform to type I or II , class 1 or 2, ticking of CCC-C-436, style a or b (see 6.2).

3.3.7 Sheeting. The cotton sheeting shall conform to type VII or VIII, class 1 of CCC-C-432, or type I or II, class 1 or 2, sheeting of CCC-C-430, style D or E.

3.3.8 Open mesh or nonwoven fabric. The open mesh fabric shall have a weight of 1-1/2 ounces to 3-1/2 ounces per square yard and shall be made from a thermoplastic material. The nonwoven fabric shall have a weight of 1-1/2 ounces to 3-1/2 ounces per square yard and shall be either a spun bonded nylon, polyester or polypropylene fabric or needle-punched thermally bonded polyprolylene fabric (see 3.5.11.1 and 4.4.1.2).

3.3.9 Heat perimeter shield or flange. The material used for flanging or a heat perimeter shield shall be an aluminized fabric (see 4.4.1.2).

3.3.10 Thread. The thread used for stitching the edge binding tape and the border shall be cotton, ticket number 30, 3 ply, type IA1 or IA2 of V-T-276 or the thread shall be polyester, type I, class 1, subclass A, size E to FF of V-T-285 or the thread shall be nylon, type I or type II, class A, size E to FF of V-T-295. The thread used for quilting shall be polyester conforming to type I or II, size B, class 1, subclass A of V-T-285.

3.3.11 Tape.

3.3.11.1 Edge binding tape. The tape used in finishing the edges of the mattress shall be 5/8-inch (minimum) width polyester or polyester warp yarn and polypropylene filling yarn tape (see 6.7); or an approved fire resistant or smoldering ignition resistant treated cotton tape (see 6.5). The polyester or polyester/polypropylene tape shall have a minimum breaking strength of 100 pounds and the treated cotton tape shall have a minimum breaking strength of 30 pounds. The color of the tape shall be white or natural. Testing shall be as specified in 4.4.1.1.

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3.3.11.2 Tufting tape, braid or twine. The tape braid or twine used for tufting shall be either nylon or polyester. The nylon or polyester tape shall be not more than 5/8 inch wide and shall have a 50 pound minimum breaking strength. The breaking strength for the nylon or polyester tufting braid or twine shall be 80 pounds minimum. In addition to nylon or polyester tape, innertufted mattresses can use cotton tape, braid, or twine for tufting. The cotton tape shall be not less than 3/8 inch nor more than 5/8 inch wide and shall have a minimum breaking strength of 50 pounds. The cotton tufting braid or twine shall have a breaking strength of not less than 32 pounds. Testing shall be as specified in 4.4.1.1.

3.3.12 Handles.

3.3.12.1 Class 1. Class 1 mattresses shall be furnished without handles.

3.3.12.2 Class 2. Class 2 mattresses shall be furnished with two horizontal handles (lifting straps) attached to the border on each side of the mattress. The handles shall be a commercial braid covered, cotton or synthetic cord lifting strap utilizing nonferrous eyelets or grommets, metal clip cord ends and a spring steel reinforcing strip for each handle. The diameter of the braid covered cord and the finished length of the handles and the spacing of the grommets or eyelets shall be consistent with good commercial practices (see 4.4.1.2).

3.3.13 Unicellular polyethylene foam. The border of the mattress shall be constructed by using unicellular polyethylene foam. The foam shall be 1/4 $\pm$ 1/32 inch thick and 5 $\pm$ 1/4 inch wide and have a density of 1.6 to 2.4 pounds per cubic foot. The surface of the foam material shall show no cracks, tears, or separations resulting from bending around a mandrel having a diameter of four times the thickness of the foam. The foam material shall not support active mold growth and shall not be treated with fungicide. The polyethylene foam shall be free from any objectionable odor.

3.3.14 Tufting buttons. The tufting buttons shall be a commercial type of the size, strength, shape, and finish commonly used in mattress construction. The color of the tufting buttons shall be either white, blue, or egg-shell (grayish white). All tufting buttons contained in a mattress shall be of the same color i.e., all white, all blue, or all egg-shell (see 4.4.1.2) (for tufting see 3.5.11).

3.3.15 Ventilators. The ventilators shall be a good commercial type constructed of brass, aluminum, or 400 series steel and shall not rust. The length of the ventilator shank shall be of a sufficient length to effect a secure clinch of the border without cutting the surrounding ticking. A minimum of 50 percent of the face shall be perforated.

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3.3.16 Wire mat. When wire mat is used as the spring covering specified in 3.5.5, it shall conform to the following: The wire mat shall have a series of steel wires spaced 1 inch ± 3/8 inch apart, running at right angles to longitudinal fiber cords which are uniformly spaced not more than 8 inches apart. The steel wires shall pierce and pass through the center cords and shall be crimped around the border cords. The wire shall be 19-gage oil-tempered steel wire with a tensile strength range of 266,000 psi to 296,000 psi. The spirally wound fiber cords shall be constructed of not less than 30 pound kraft paper. The center cords shall be not less than 0.16 inch in diameter, with or without a core, and shall have a breaking strength of not less than 100 pounds. The border cords shall be not less than 0.125 inch in diameter with a breaking strength of not less than 100 pounds. The dimensions of the wire mat shall be 3 inches ± 1/4 inch in length and 2 inches ± 1/4 inch less in width of the dimensions of the applicable spring unit used for the end item mattress. Inspection of the completely fabricated wire mat shall be as specified in 4.4.2.2.

3.3.17 Hog rings. The hog rings (see 3.5.5) shall be commercial type, of the size, strength, and finish commonly used in mattress construction (see 4.4.1.2).

3.3.18 Coils.

3.3.18.1 Coils for type II mattress. The wire for all coils shall be hard-drawn, spring steel wire. The wire for the edge and corner coils shall be not less than 13 gage wire having a minimum tensile strength of 205,000 psi and a maximum of 250,000 psi. The wire for all other coils for regular firmness shall be of 13-1/2 to 14 gage wire having a minimum tensile strength of 212,500 psi and a maximum of 260,000 psi, coil wire for firm units shall be of 13 to 13-1/2 gage having a minimum tensile strength of 205,000 psi and a maximum of 275,500 psi, and coil wire for extra firm units shall be of 12-1/2 to 13 gage having a minimum tensile strength of 205,000 psi and a maximum of 275,500 psi. Corner coils shall be cylindrical, with not less than 6 convolutions. The ends of the corner coils shall be die formed to 2-1/2 to 4-1/2 inch radius. End coils shall be open end offset with sufficient convolution to be consistent with the inside coils. Edge coils forming the sides of the spring unit shall present a straight edge to the outer edge of the spring unit and shall be cylindrical. All coils except corner coils shall be hour-glass shaped and shall have two precision offset sections in the top and bottom convolution of the coil, die formed to accurately place the coil in its proper transverse position to receive the interlocking helical. Helicals surround each offset portion of the coils and securely hold each coil in position to permit complete freedom of hinging action. All other inside coils shall be hour-glass shape. All coils except the corner coils shall have not less than 4-1/2 active turns or complete convolutions. All coils shall be free-end type coils to allow their being anchored within the helical without accessory locking device. The coils shall be 5 inches ± 1/4 inch in height. Coils or the complete spring unit assembly shall be given suitable thermal treatment to relieve residual stresses caused by coiling (see 4.4.1.2). Coils shall not decrease in height more than 3/16 inch when tested as specified in 4.4.1.1.

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3.3.18.2 Coils for type III mattress. The wire for the spring coils shall be hard-drawn, spring wire. Coil wire for regular firmness shall be of 13-1/2 to 14 gage wire having a minimum tensile strength of 212,500 psi and a maximum of 260,000 psi. Coil wire for firm units shall be of 13 to 13-1/2 gage wire having a minimum tensile strength of 205,000 psi and a maximum of 275,500 psi. Coil wire for the extra firm unit shall be of 12-1/2 to 13 gage wire having a minimum tensile strength of 205,000 psi and a maximum tensile strength of 275,500 psi. All coils shall be hourglass in shape, except for the coils in the two longitudinal rows forming the sides of the spring unit which may be cylindrical or elliptical in shape. There shall be not less than three active turns or convolutions per coil, including only portions between the knots of the coil, and the coils shall be 5 inches ± 1/4 inch in height. Coils or the complete spring unit assembly shall be given suitable thermal treatment to relieve stresses caused by coiling (see 4.4.1.2). Coils shall not decrease in height more than 3/16 inch when tested as specified in 4.4.1.1.

3.3.19 Innerspring units.

3.3.19.1 Type II mattress (all-wire spring unit, free-end coil construction). The spring unit shall be made with materials specified in 3.3.18.1, 3.3.20, 3.3.21 and 3.3.22 located so as to keep edges from sagging, prevent sidesway, to give firm support to all parts of the body, and to maintain flexibility. All-wire springs units, free-end coil construction, shall consist of transverse rows of alternating right-hand and left-hand coils connected with a long helical specified in 3.3.21 running at right angles to the long axis of the unit in such a manner as to prevent lateral slippage. The free end of the coil shall be anchored within the helical without the use of any accessory locking device. The edges of all outside coils shall be constructed so as to present a straight edge to the construction. The end rows shall have a stabilizer wire specified in 3.3.20 placed within each of the 4 end helicals with the free ends of wire extending to a point on line with the center of the second row of coils at each end, and fastened so as to prevent slippage when tested as specified in 4.4.6 for the folding characteristic. The 4 free-end corner coils of the unit construction shall be cylindrical. The ends of the helicals shall be turned in to form a smooth, rounded termination. The unit shall meet the firmness requirements as specified in 3.3.23 and 6.2.

3.3.19.2 Type III mattress (all-wire spring unit, knotted coil construction). All-wire spring units, knotted-coil construction, shall consist of transverse rows of coils specified in 3.3.18.2, 3.3.20, 3.3.21 and 3.3.22 connected by helicals running at right angles to the long axis of the unit to prevent lateral slippage and to permit a free-hinge action when the mattress is folded or rolled from end to end; or shall consist of transverse rows of individual coils or units of not more than 4 helical convolutions per coil, joined to the neighboring coils or units of coils to prevent lateral slippage and to permit a free-hinge action when the mattress is folded from end to end when tested as specified in 4.4.6. Where helicals are used in the construction, the individual

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spring coils may be offset with a portion or portions passing through not less than 3 turns of the helical. All knots shall be turned down and the ends of all helicals shall be turned inward forming a smooth and rounded termination. The unit shall meet the firmness requirements as specified in 3.3.23 and 6.2.

3.3.20 Stabilizer wire (end rods). The stabilizer wire shall be made from high carbon upholstery steel wire of 9 to 13 gage having a minimum tensile strength of 165,000 psi and a maximum of 225,000 psi when tested as specified in 4.4.1.2. The corners shall be die formed to a 2-1/2 to 4-1/2 inch radius. The stabilizers shall be attached to the outside of each end row of coil so as to prevent slippage, and shall be formed to fit the full width of the spring unit and extend around the corners of the unit to the center of the second row or coils and fastened to prevent slippage.

3.3.21 Cross helicals. The helical wire shall be made from hard-drawn high-carbon steel lacing wire, not less than 17-1/2 gage. The formed helical shall be of sufficient diameter to connect the coils, and shall have a 15/32 inch pitch or less. The helical shall have not less than 24 turns per foot. All helicals shall be given suitable thermal treatment to relieve residual stresses caused by coiling (see 4.4.1.2.).

3.3.22 Helicals for attaching stabilizers (end rods). The helicals shall be made from high-carbon upholstery steel not less than 18 gage, with not less than 24 turns per foot. The formed helical shall have an outside diameter sufficient to connect the stabilizer wire and coils and shall have a 15/32 inch pitch or less. All helicals shall be given a suitable thermal treatment to relieve residual stresses caused by coiling (see 4.4.1.2).

3.3.23 Firmness for mattress spring units. There shall be three firmnesses for mattress spring units as determined by a deflection test (see 4.5.3). The load versus the deflection for each firmness shall be as listed in table I.

TABLE I. Load versus deflection

Deflection (inches)	Loading (pounds)		
	Regular	Firm	Extra firm
1	22-33	35-50	50-70
1-1/2	38-52	60-75	75-100
2	55-70	83-100	110-135

3.4 Design. Mattresses shall be either outertufted, innertufted or quilted.

3.5 Construction. The construction requirements for all types of mattresses shall be as specified herein.

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3.5.1 Stitching. Stitching shall conform to stitch type 301 or 401 of FED-STD-751 with 8 to 12 stitches per inch (except for quilting). When 401 stitching is used, the chain portion of the stitching shall not appear on the outside of the ticking.

3.5.2 Seams. When longitudinal pieces of ticking are used, they shall be stitched with stitch type 401 and seamed with seam type LSb-2 or stitch type 301 and seam type SSa-1 conforming to FED-STD-751.

3.5.3 Type 301 stitching. Ends of all seams and stitching when not caught in other seams and stitching shall be backstitched or overstitched 3/4 inch minimum, except for closing the mattress on a tape edge machine. Thread tension shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.5.3.1 Repairs of type 301 stitching. When making the repairs listed in a. and b. below, the ends of stitching are not required to be backstitched. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 3/4 inch back of the end of the stitching.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 3/4 inch in back of the defective area, continue over the defective area, and continue a minimum of 3/4 inch beyond the defective area onto the existing stitching. Loose or excess-ively tight stitching shall be repaired by removing the defective stitching, without damaging the materials, and restitching in the required manner.

3.5.4 Types 401, 504, and 505 stitching. Thread tension shall be maintained so that there will be no loose stitching. All repairs shall be in accordance with 3.5.3.1(a) and 3.5.3.1(b) and type 301 stitching may be used for making the repairs of type 401 stitching.

3.5.5 Insulation for inner spring unit. The innerspring units of mattresses shall be covered by either sheeting specified in 3.3.7 or by the open-mesh or nonwoven fabric specified in 3.3.8 or by the wire mat specified in 3.3.16. The sheeting or nonwoven fabric shall be attached to both the top and bottom surfaces of the innerspring unit in accordance with the contractor's commercial practice. The wire mat shall be securely fastened to the innerspring unit by three commercial hog rings attached along each longitudinal side. Two of the hog rings shall be located near the ends of the spring unit. Twelve hog rings (total) are required to fasten the wire mat to the spring unit. The hog rings shall not be clamped around the fiber cord on the edge of the wire mat, but shall be clamped around one wire of the mat and the top of the border coil.

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3.5.6 Felt padding for spring units. The springs and insulation shall be padded with felt specified in 3.3.4. The felt shall be placed over the spring unit and cover assembly in full length and full width layers on both surfaces of the mattress. The felt shall be distributed in layers of equal weight and uniform thickness to adequately cover all areas including edges, ends, and corners. A special crown of felt consisting 5 to 15 percent of the total weight of felt on each face shall be uniformly distributed over the middle 1/3 of the length face and for the full width of the mattress for the grade A and grade B felts only. The corners shall be well padded and firm. Both sides of the mattress shall be finished so that either side may be used as the sleeping side with equal comfort.

3.5.7 Barrier material (toppers). The contractor shall use a barrier system or an approved flame retardant or smoldering and ignition resistant treatment of the felt to meet the mattress flammability provisions specified in 3.2. When a barrier system is used, the barrier materials specified in 3.3.5 shall be placed directly over the felt padding. Barrier materials may also be quilted to the ticking fabric for quilted top mattresses (see 3.5.12). Innertufted mattresses may use a tufting support fabric (See 3.3.8) over the barrier materials. The use of a tufting support fabric (see 3.3.8) for outer tufted mattresses is optional. When approved treatments are to be applied to the felt padding the use of a barrier system or tufting support fabric is optional (see 3.3.5.3).

3.5.8 Cover construction. The top and bottom sections in the mattress cover shall each be made up of not more than two longitudinal pieces of ticking specified in 3.3.6 when style a ticking is specified, it shall be matched and sewn together (see 3.5.2). Furthermore, the blue stripes of the ticking shall run the lengthwise direction of the mattress. When flanging is not used or when a heat perimeter shield is not used, the top and bottom panels shall have hemmed edges.

3.5.8.1 Flange or heat perimeter shield attachment. When a flange or heat perimeter shield is used, it shall be 1-3/8 inches to 5-1/2 inches wide and shall be the aluminized fabric specified in 3.3.9. The flange or heat perimeter shield shall be attached to the ticking by using one of the following operations:

a. The flange or heat perimeter shield shall be attached by overedge stitching it to the raw edge of the ticking.

b. The flange or heat perimeter shield shall be superimposed on the ticking and seaming with a row of stitches 1/16 to 1/8 inches from the edge.

c. The ticking shall be prehemmed and the flange or heat perimeter shield on the ticking shall be seamed with a row of stitches 1/16 to 1/8 inch from the edge; the operations may be accomplished simultaneously.

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The flange or heat perimeter shield shall also be stitched to the ticking with a second row of seaming stitches (stitch type 301 or 401). This row of stitching shall be positioned along the opposite long edge of the flange or heat perimeter shield material and shall be sewn a minimum of 1/4 inch from the edge. Both rows of stitching of the heat perimeter shield or flange, after overstitching or backstitching the ends of stitching, may be carried off the side edge of the ticking cover and then trimmed. When 401 stitching is used, the chain portion of the stitching shall not appear on the outside of the ticking. When overedge stitching is used it shall be accomplished with stitch type 504 or 505, 3/16 inch gage of throw, 5 to 7 stitches per inch. The second row of seaming stitches shall have 5 to 7 stitches per inch. Thread shall be as specified in 3.3.10.

3.5.9 Border construction. The border shall be prebuilt prior to attachment to cover. The padding for the border shall be either the felt specified in 3.3.4 or polyester batting specified in 3.3.5 or the unicellular polyethylene foam specified in 3.3.13. The use of other approved barrier materials (see 3.3.5.2) as padding in the construction of the border is also allowed. The border shall be constructed by using the ticking specified in 3.3.6 along with one of the padding materials specified above. The border may be constructed with or without the use of a backing material, as indicated below. The outside of the border shall be made of ticking with both edges hemmed with machine stitching type EFa-1, with a minimum of 1/4 inch hem; or the ticking, padding and with or without a backing material may be sewn together to join the border in one operation. The stitching shall run perpendicular to one folded edge of ticking, run parallel across to form a hem approximately 1-5/8 inches, and run perpendicular to the opposite folded edge of ticking, going parallel again to form a hem approximately 1-5/8 inches. This operation will result in an alternating stitch and skipped hemmed edge along both sides for the entire length of the mattress border. The border on each mattress shall be made of not more than two pieces of ticking with the ends spliced and sewn together with seam type SSa-1, in a continuous manner so as to have no break in its uniformity and finish. The ticking shall be padded with padding material and the width of padding shall be the width of the hemmed border plus 1/4 inch or minus 1/2 inch. The padding shall be butted or spliced at the seam joining the ends of the border in such a manner as to leave no breaks in its uniformity and finish. A backing material shall be required when felt paddings specified in 3.3.4.1 and 3.3.4.2 are employed. Backing materials are optional when borders are constructed with polyester batting or unicellular polyethylene foam padding. When a backing material is used on the border, the material shall either be sheeting specified in 3.3.7 or ticking specified in 3.3.6 or the nonwoven fabric specified in 3.3.8 or the aluminized flange and heat shield fabric specified in 3.3.9. The width of the backing material shall be at the option of the contractor. The border shall be sewn together with not less than six stitches per inch. The border shall be equipped with ventilators specified in 3.3.15 and in table III and IV. The ventilators shall be placed in a line approximately half way between the top and bottom hem of the border and shall be secured to border at reasonably spaced intervals (approximately equidistant) on the long-

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itudinal sides of the border. Ventilators shall pass completely through the entire border from the outside. When class 2 mattresses are specified, the border shall be equipped with two horizontal handles specified in 3.3.12 on each side of the mattress border and located approximately one third of the length from each end of the mattress. The felt padding specified in 3.3.4.1 shall weigh not less than 3.0 ounces per linear yard (see 4.4.1.2). The polyester batting padding specified in 3.3.5 shall weigh not less than 10 ounces per square yard when used in borders (see 4.4.1.2). The entire border shall be constructed so as not to interfere with the free passage of air through the ventilators.

3.5.10 Edge construction. The finished edge of the cover ticking and the finished edge of the border shall be joined together and sewn within the folds of the edge binding tape specified in 3.3.11.1. The tape shall extend completely around both edges of the mattress. Not more than three pieces of tape shall be used on each border edge not including normal bias splicing.

3.5.11 Tufting. Mattresses shall be designed to be either outertufted or innertufted. Tufting shall be accomplished using either button tufts specified in 3.3.14 or tufting twine, braid, or tape specified in 3.3.11.2. The depth of tufts shall be 4-1/2 inches to 5-1/2 inches. The number of tufts and the pattern shall be as specified in table II. Mattresses shall be tufted through both sides. Knots of tufts shall be arranged so that they will not appear above the surface of either side of the mattress.

3.5.11.1 Innertufted mattresses. Mattresses may be innertufted, resulting in a smooth surface mattress with concealed innertufting. The tufting requirement specified in 3.5.11 shall also apply to innertufted mattresses. Innertufted mattresses may require the use of a tufting support fabric to properly anchor the filling materials. The tufting support fabric shall be as specified in 3.3.8. Any tear, rip, or cut in the tufting support fabric shall be repaired prior to filling the mattress in the ticking cover. The number of tufts, pattern, and spacing requirements are outlined in table II.

TABLE II. Tufting requirements

Size	Number of tufts 1/	Pattern 2/
1	17	2x6
2	28	3x6
3	28	3x6
4	28	3x6
5	28	3x6
6	28	3x6

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TABLE II. Tufting requirements (cont'd)

Size	Number of tufts 1/	Pattern 2/
7	28	3x6
8	28	3x6
9	39	4x6
10	39	4x6
11	39	4x6
12	39	4x6
13	28	3x6
14	28	3x6
15	39	4x6
16	39	4x6

1/ The spacing of the tufts shall be uniform. For mattresses of 80 inches or longer, there shall be an additional tuft in each lengthwise row of every 10 inch increase in length.

2/ The arrangement of the tufts shall be in a diamond pattern. The distance of any row closest to an edge shall be 6 inches plus or minus 1 inch and the distance of any row closest to an end shall be 6 inches minus 1 inch, plus 2 inches.

3.5.12 Quilting. Mattresses can be designed to have a quilted top and a bottom surface. The mattress shall be quilted using a commercial quilt pattern. The quilted surface mattress shall be furnished without fabric splices (joining of ticking outer covering) and without areas evidencing complete bobbin changes and shall be consistent with good commercial quality. The quilting shall have 5 to 9 stitches per inch. The filling materials of type II quilted surface mattress shall be anchored by inner tufting. The innertufted requirements specified in 3.5.11 shall also be applicable for type II quilted surface mattresses. The filling materials of type III quilted surface mattress shall be anchored in accordance with good commercial practices.

3.5.13 Corners. The four corners of the mattresses shall be firm and well padded. The four corners of each mattress shall be rounded conforming to the configuration of the spring unit.

3.6 Physical requirements. Finished mattresses shall conform to the requirements specified in tables III and IV.

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TABLE III. Finished physical requirements - Type II

Size	Length (inches) ± 1/2 inch 3/	Width (inches) ± 1/2 inch 3/	Thickness (inches) ± 1/2 inch	Weight (pounds) min 1/	Number of ventilators on each side	Total No. of coils (min)			Weight of felt layer on each face (pounds) 1/
						Regular	Firm	Extra firm	
1	76	29	6-1/2	30	4	133	154	161	8-1/2
2	78	33	6-1/2	39	4	152	184	192	10
3A 2/	75	35	-	-	-	-	-	-	-
4A 2/	74-1/2	36	6-1/2	40	4	162	198	207	10/12
5	78	36	6-1/2	41	4	171	207	216	11
6	84	36	6-1/2	47	4	180	216	225	12
6A 2/	84	35	-	-	-	-	-	-	-
7 2/	74-1/2	38	6-1/2	45	4	162	198	230	11-1/2
8	78	39	6-1/2	47	4	171	207	240	12
9	78	41-1/2	6-1/2	50	4	190	230	264	13
10	78	47	6-1/2	57	4	209	253	288	15
11	73	53	6-1/2	58	4	234	286	322	16
12 2/	74-1/2	53	6-1/2	60	4	234	286	322	16
13 2/	79-1/2	36	6-1/2	44	4	171	207	216	11
13A 2/	80	35	-	-	-	-	-	-	-
14 2/	79-1/2	38	6-1/2	48	4	171	207	240	12-1/2
15 2/	79-1/2	53	6-1/2	65	4	247	299	336	17-1/2
16 2/	79-1/2	60	6-1/2	65	4	285	345	384	17-1/2

1/ See 6.9 for innerspring unit dimensions information.

2/ Not a specification requirement, included for information purposes.

3/ These sizes are the same as in the proposed Voluntary Product Standard TS 222 (see 6.8).

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TABLE IV. Finished physical requirements - Type III

Size	Length (inches) ± 1/2 inch 3/	Width (inches) ± 1/2 inch 3/	Thickness (inches) ± 1/2 inch	Weight (pounds) min 1/	Number of ventilators on each side	Total No. of coils (min)			Weight of felt layer on each face (pounds) 1/
						Regular	Firm	Extra firm	
1	76	29	6-1/2	30	4	126	126	168	8-1/2
2	78	33	6-1/2	30	4	154	154	154	10
3A 2/	75	35	-	-	-	-	-	-	-
4 2/	74-1/2	36	6-1/2	36	4	147	147	192	10-1/2
5	78	36	6-1/2	39	4	154	154	154	11
6	84	36	6-1/2	41	4	184	184	216	12
6A 2/	84	35	-	-	-	-	-	-	-
7 2/	74-1/2	38	6-1/2	43	4	168	168	216	11-1/2
8	78	39	6-1/2	41	4	176	176	225	12
9	78	41-1/2	6-1/2	43	4	198	198	250	13
10	78	47	6-1/2	50	4	220	220	275	15
11	73	53	6-1/2	51	4	252	252	308	16
12 2/	74-1/2	53	6-1/2	60	4	252	252	308	16
13 2/	79-1/2	36	6-1/2	36	4	176	176	208	11
13A 2/	80	35	-	-	-	-	-	-	-
14 2/	79-1/2	38	6-1/2	40	4	176	176	216	12-1/2
15 2/	79-1/2	53	6-1/2	60	4	264	264	264	17-1/2
16 2/	79-1/2	60	6-1/2	60	4	286	286	384	17-1/2

1/ Not a specification requirement, included for information purposes.

2/ These sizes are the same as in the proposed Voluntary Product Standard TS 222 (see 6.8).

3/ See 6.9 for innerspring unit dimensions information.

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3.7 Labels. All labels shall be sewn at one end of the mattress into the same edge seam.

3.7.1 Statutory label. Statutory labels shall conform to the legal requirements of the state in which the mattresses are manufactured.

3.7.2 Identification label. Each mattress shall have an identification label conforming to type VI, class 5 of DDD-L-20, with the following exceptions:

a. The requirement for launderability shall not apply.

b. The spring unit supplier's name and type of spring unit along with the lot identification number (flammability) shall be included in the marking.

c. The identification label bearing all the required markings shall also have the degree of firmness included in the marking. The degree of firmness shall be stamped in 1/2 inch (minimum) capital letters as follows:

REG
FIRM
X-FIRM

The regular (REG) shall be stamped with black ink, the firm (FIRM) in green ink and the extra firm (X-FIRM) in red ink. The identification label shall be attached to each mattress. Statutory and medical labels (when required) together with an identification label shall be sewn to the mattress.

3.7.3 Medical label. When specified (see 6.2), mattresses procured for medical activities shall have a label conforming to type VI, class 9 of DDD-L-20, bearing manufacturer's name or registered trade mark, FSN, item description and lot identification number (flammability).

3.8 Workmanship. The finished mattresses shall conform to the quality and grade of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1), it shall be examined for the defects specified in 4.4.4.1.1 and 4.4.4.1.2 and tested for the characteristics specified in 4.4.6. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1.1 Testing of components. In addition to the quality assurance provisions of the referenced specifications, components and materials listed in table VI shall be tested for the characteristics shown therein. The methods of testing specified in FED-STD-191, wherever applicable, and as specified herein shall be followed. When the data in the "Number of determinations per sample unit" and the "Results reported as" columns are not specified in table VI, they shall be as required in the referenced test method. All test reports shall contain the individual values utilized in expressing the final result. The sample size shall be in accordance with table V and the lot shall be unacceptable if one or more sample units fail to meet any requirement specified.

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TABLE V. Sampling for testing

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

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TABLE VI. Test methods of components

Component and lot expressed in terms of	Characteristic	Requirement paragraph	Test method	Number determinations per sample unit	Results reported as	Sample unit
Edge binding tape (1 spool)	Material identification	3.3.11.1	Standard commercial	1	Pass or fail	
	Breaking strength	3.3.11.1	6016 1/ 2/	-	-	1 cone, spool or 30 feet of tape
	Width	3.3.11.1	Rule	3	Pass or fail	
Tufting tape, braid or twine (1 cone or spool)	Material identification	3.3.11.2	Standard commercial	1	Pass or fail	
	Breaking strength	3.3.11.2	6015 1/ 2/	-	-	1 cone, spool or 30 feet of tape
	Width (when applicable)	3.3.11.2	Rule	3	Pass or fail	
Coils, mattresses 100 coils or 1 complete spring unit assembly	Compression Set	3.3.18.1 or 3.3.18.2	4.5.1 3/	3	Pass or fail	2 coils of each type or 1 complete spring unit assembly (as applicable)
Spring unit (1 complete unupholstered spring unit)	Firmness	3.3.23	4.5.3 3/	1	Pass or fail	1 complete spring unit assembly

1/ Refers to FED-STD-191.

2/ Except that the full width of the tape shall be used.

3/ A contractor's certificate of compliance shall be furnished and will be acceptable for the stated requirements.

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4.4.1.2 Components, materials, and component assemblies. When the components, materials, and component assemblies listed in table VII are used in the fabrication of mattresses, a contractor's certification of compliance (see 4.1.2) shall be furnished and will be acceptable for the specified requirements.

TABLE VII. Assembly requirements

Component, material or component assembly	Requirement paragraph
Cotton 1/	3.3.1
Linters 1/	3.3.2
Polyester fiber	3.3.3
Felt 1/	3.3.4
Grade of felt	3.3.4
Barrier materials	3.3.5
Inner support tufting fabric	3.3.8
Heat shield or flange fabric	3.3.9
Handles	3.3.12.2
Unicellular polyethylene foam	3.3.13
Tufting buttons	3.3.14
Ventilators	3.3.15
Wire mat components (wire and fiber cords)	3.3.16
Hog rings	3.3.17
Coils	3.3.18
Stabilizer wire (end rods)	3.3.20
Helicals	3.3.21
Helical for end rod attachment	3.3.22
Firmness of mattress spring unit	3.3.23
Weight of padding on order	3.5.9

1/ The cotton or linters or felt shall be examined, to determine conformance with 3.3.1, 3.3.2 and 3.3.4 inclusive, by classifiers licensed by the U.S. Department of Agriculture; or representative samples, each weighing not less than 1/2 pound, of the cotton or linters and a minimum 12 inches by 12 inches of felt shall be forwarded to the administrative office concerned at Washington, DC where inspection by the proper board of the U.S. Department of Agriculture shall be obtained. The opinion of this board as to the true grade or quality will be final. Inspection reports from the USDA shall be furnished and will be acceptable for the stated requirements.

4.4.2 Innerspring and wire mat inspection. Intermediate inspection shall be performed on all innerspring units, except when units are listed on an ASL (Acceptable Supplier's List). Inspection shall also be performed on the wire mat when used as an insulator. The defects found during examination shall be in accordance with 4.4.2.1, 4.4.2.2 and 4.4.2.3. The inspection level shall be S-3 and the acceptable quality level (AQL) shall be 4.0 defects per class, expressed in defects per hundred units.

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4.4.2.1 Visual examination of all-wire spring unit, open end offset coil construction assembly (type II mattress). The sample unit for this examination shall be one type II mattress spring unit. Examination for defects shall be as listed in table VIII.

TABLE VIII. Type II assembly visual defects

Examine	Defect
Construction	Transverse rows do not consist of alternating right-hand and left-hand coils
	Alternating right-hand and left-hand coils are not connected with a helical running at right angles to the long axis of the unit in such a manner as to prevent lateral slippage
	Free end of coil is not anchored within the helical
	Intermediate convolutions of the individual coils in the longitudinal rows forming the side are not cylindrical in shape
	Edges of all outside coils are not constructed so as to present a straight edge to the construction
	The individual coils forming the end row contain less than 5 convolutions
	Corners of stabilizer wires are die-formed to less than 2-1/2 inch or more than 4-1/2 inch radius
	The free end of the stabilizer wire is not placed within each of the 4 end helicals and the free ends of wire do not extend to a point on line with the center of the second row of coils at each end and not fastened so as to prevent slippage
	Less than 6 convolutions on any free-end corner coil
	Any free-end corner coil not cylindrical in shape
	Any corner coil formed to less than 2-1/2 inch or more than 4-1/2 inch radius
	Not an hourglass shape on any other coil not indicated above
	Coil not indicated above having less than 4-1/2 active turns or complete convolutions (when counted from last point of contact of coil with helical on top to the first point of contact with the helical on the bottom)
	Coil having a height less than 4-3/4 inches or more than 5-1/4 inches
	Total number of coils less than specified
	End of helicals not turned in forming a smooth rounded termination

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4.4.2.2 Visual examination of all-wire spring unit, knotted-coil construction (type III mattress). The sample unit for this examination shall be one spring unit as applicable for construction type for type III mattress. Examination of the spring unit for defects in material design and workmanship for the applicable construction type shall be as specified in table IX.

TABLE IX. Type III construction visual defects

Examine	Defect
Construction	Spring unit does not consist of either transverse rows of coils connected by the use of helicals running at right angles to the long axis of the unit as specified; or the spring unit does not contain transverse rows of individual coils, or units of not more than 4 helical convolutions per coil to a unit joined to the neighboring coils or units as specified
	Coil contains less than 3 active turns or convolutions (including only portions between knots of the coil)
	Coil height is less than 4-3/4 inches or more than 5-1/4 inches
	Corners of stabilizer wires are die-formed to less than 2-1/2 inches or more than 4-1/2 inches radius
	Total number of coils less than specified
	All knots not turned down
	Any end of any helical not turned in so as to form a smooth and rounded termination of the helicals

4.4.2.3 Dimensional examination of wire mat. The sample unit for this examination shall be one completely fabricated wire mat. Defects shall be as listed in table X.

TABLE X. Wire mat dimensional defects

Examine	Defect
Dimensions (length and width)	Not as specified
Spacing of steel wires	Not as specified
Spacing of fiber cords	Not as specified

4.4.3 In-process inspection. Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplishment of the items listed in table XI is in accordance with the specification requirements.

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TABLE XI. In-process inspection

Operation or assembly	Requirement paragraph
Application of insulation on innerspring units	3.5.5
Application of padding (upholstery) to innerspring units	3.5.6
Application of barrier material (toppers) to spring units (when applicable)	3.5.7
Construction of cover (with or without heat shield or flange)	3.5.8
Construction of border	3.5.9
Construction of edges	3.5.10
Concealed operations of innertufted mattresses	3.5.11, 3.5.11.1 and 3.5.12

4.4.4 Inspection of the end item.

4.4.4.1 Visual examination of mattresses. The defects found during examination shall be classified in accordance with 4.4.4.1.1 and 4.4.4.1.2. The sample unit for these examinations shall be one mattress (as applicable). Any mattress removed from the polyethylene bag for examination need not be resealed. The open end of the polyethylene bag shall be repaired (taped up) prior to placing the packaged mattress (mattress in bag) back into a carton.

4.4.4.1.1 Visual examination of mattress. The mattress shall be examined for defects in design, material, construction, and workmanship as listed in table XII. The inspection level shall be II, and the AQL shall be 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

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TABLE XII. Mattresses visual defects

Examine	Defect	Classification Major	Minor
Quality of ticking	Hole, cut, tear, smash, mend, or open place:		
	Up to 1 inch, inclusive		X
	More than 1 inch	X	
	Loose yarn, shagged thread, misweave, missing yarn, broken yarn, thin place, shade bar, or woven in waste:		
	Up to 1 inch, inclusive		X
	More than 1 inch	X	
	Rancid or other objectionable odor	X	
	Not fabric specified	X	
	Start marks (heavy or light)		X
	Spot or stain 1/4 inch or over in diameter, clearly noticeable when viewed at a distance within 3 feet		X
	Excessive neppiness	X	
Seams and stitching	Any required seam or stitching operation omitted	X	
	Ticking not seamed with seam type SSa-1 or LSb-2		X
	Not stitch type 301 or 401 (as applicable)		X
	Stitches per inch less than specified	X	
	Stitches per inch more than specified		X
	Ends of any seam not backstitched (except when end of seam is caught in another seam or stitching, or when closing the mattress) as specified		X
	Raw edge more than 1 inch in length		X
	Open seam more than 1 inch in length	X	
	Open seam 1 inch or less in length		X

NOTE: A seam shall be classified as open when one or more stitches are broken or when two or more continuous skipped or run-off stitches occur.

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TABLE XII. Mattresses visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Seams and stitching (cont'd)	Stitch tension loose, resulting in a loosely secured seam		X
	Stitch tension tight (stitches break when normal strain is applied to the seam or stitching)		X
	NOTE: Puckering is evidence of tight tension, when puckering is evident, the inspector shall test seams by exerting pull on seam.		
	Needle chew, hole, or cut:		
	Up to 1 inch, inclusive		X
	More than 1 inch	X	
	Seam badly puckered, pleated, twisted or caught in unrelated operation		X
	Five or more untrimmed thread ends or loose threads, 1 inch or more in length		X
Construction	Not type specified	X	
	Sagging appearance, i.e., filling not evenly distributed	X	
	Lumps clearly noticeable	X	
	Thin hollow places clearly noticeable	X	
	Border or corner contains insufficient padding	X	
	Spring coils can be felt through ticking	X	
	Face-panel (top or bottom) ticking more than two pieces of material	X	
	Style a blue stripe of ticking does not run in lengthwise direction		X
	Style a ticking stripes not properly matched adversely affecting appearance		X
	Corner(s) not firm	X	
Border and bound edges	Border not constructed as specified	X	
	Border fabricated with more than two pieces of ticking (all around)		X
	Transverse stitching on border does not extend across the full width of the border		X
	Transverse stitching on border less than 1-1/4 inch on centers or more than 1-3/4 inches on centers	X	

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TABLE XII. Mattresses visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Border and bound edge (cont'd)	Not box edge or box edge not sharply defined		X
	Binding tape does not extend completely around edge of mattress	X	
	More than three pieces of binding tape on edge of mattress	X	
	Binding tape puckered, pleated, or twisted		X
	Color of tape not as specified		X
	Ventilators insecurely attached or not firmly anchored to border		X
	Less than four ventilators in each side	X	
	Evidence that ventilator passage is obstructed to an extent interfering with free passage of air into the spring section		X
	Any ventilator not aligned halfway between the top and bottom of border (permit ± 1/2 inch tolerance)		X
	Ventilators not equidistant on either side of ends of mattresses (permit ± 1/2 inch tolerance)		X
	Ventilator screen damaged		X
	Less than two handles on each side of mattress (when applicable)	X	
	Any handle misplaced or misaligned		X
	Horizontal handle pulls out of retainer (check by turning mattress over, using all handles)	X	
	Horizontal handle grommet or eyelet insecurely attached to border	X	
Tufting	Not type specified	X	
	Not accomplished as specified		X
	Tufts do not appear on both sides of mattress	X	
	Less than the number of tufts specified in table II		X
	Knots of tape tying tufts (when applicable) appear above surface on either side of mattress		X

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TABLE XII. Mattresses visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Tufting (cont'd)	Tear over 3/4 inch long in ticking under button (check underneath a minimum of three buttons on each side of mattress		X
	Arrangement of diamond pattern not as specified		X
	Spacing button tufts not uniform affecting appearance		X
	Depth of tufts not as specified		X
	NOTE: Check a minimum of three tufts by inserting a pointed thin piece of metal or wire, and of sufficient length, through the tuft until the point is felt on the bottom side of mattress. Grasp the instrument at the beginning of the depth of the top tuft and remove. Measure the distance from point of instrument to where thumb and forefinger are holding the instrument.		
Quilting	Any thin place devoid of padding 2 inches square or more in area	X	
	Any thin place devoid or padding, more than 1 inch square in area, but less than 2 inches square in area		X
	Three or more thin places devoid of padding regardless of area	X	
	Any furrow, lump, or bulge resulting from uneven or improper distribution of the padding	X	
	Five or more untrimmed thread ends or loose threads, 1 inch or more in length	X	
	Surface twisted, pleated, wavy, or puckered	X	
	A runback of more than 1-1/2 inches of broken or missing stitches in the quilting		X
	Any splice in ticking cloth	X	

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TABLE XII. Mattresses visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Quilting (cont'd)	Needle chews:		
	Up to 1 inch, inclusive		X
	More than 1 inch	X	
	Any area evidencing complete bobbin change		X
	More than 1-1/2 inches of broken or missing stitches in quilting	X	
	One or more rows of stitching omitted in quilting	X	
	Stitches per inch:		
	One or more stitches less than specified	X	
	One or more stitches more than specified		X
Labels	Missing, incomplete, incorrect, illegible, or not located as specified	X	

4.4.4.1.2 Dimensional examination. The mattress shall be examined for defects in dimensions as listed in table XIII. The inspection level shall be S-3 and the AQL shall be 4.0, expressed as defects per hundred units.

TABLE XIII. Mattress dimensional defects

Examine	Defect
Thickness	Average is less than 6 inches
	Average is more than 7 inches
Length	Average length is beyond tolerances indicated in tables III and IV
Width	Average width is beyond tolerances indicated in tables III and IV
Distance of outer tufted row to end of mattress	Less than 5 inches
	More than 8 inches
Distance of outer tufted row to edge of mattress	Less than 5 inches
	More than 7 inches

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4.4.5 Packaging inspection. An inspection shall be made in accordance with the provisions of MIL-M-3996 to determine that packaging, packing, and marking comply with the section 5 requirements of this specification.

4.4.6 Testing of the end item. Testing of the completely fabricated mattress shall be performed in accordance with table XIV for the characteristics shown therein. The sample unit shall be one completely fabricated mattress. The sample size (number of sample units) shall be in accordance with MIL-STD-105. All test reports shall contain the individual values utilized in expressing the final result.

TABLE XIV. Test methods for mattresses

Characteristics	Requirement para.	Test method	Number of determinations per sample unit	Results reported as	Insp. level	AQL
Folding of mattress freehinge action	3.3.19.1 and 3.3.19.2	4.5.2	1	Pass or fail	S-3	4.0

4.5 Test methods.

4.5.1 Coil compression or set. The height of the coil shall be measured prior to determining coil compression. Coil compression shall be accomplished by placing either end of a single coil on a flat surface and then compressing it manually, using a board or block over the top of the coil, until all adjacent convolutions are touching or the coil is in a state of total deflection. This procedure shall be repeated 20 times to determine conformity with the requirements specified in 3.3.18.1 and 3.3.18.2. If a large number of coils must be tested, a simple mechanical reciprocating type testing device may be used to accomplish the above test.

4.5.2 Folding. The mattress shall be folded together, end to end, and then opened flat again. The mattress shall be observed for lateral slippage of coils, an inability to permit a free hinge action or any visible physical distortion and thus prevent the stabilizer wire from being used all around the mattress.

4.5.3 Load deflection test. The load versus deflection test shall be conducted on an unupholstered inner spring unit. A platen 11-1/2 by 11-1/2 inches shall be placed at the center of the unit and a uniform load applied. A load sufficient to deflect the unit 2 inches shall be applied. This load shall then be released and spring unit allowed to relax for 1 minute after which the unit shall be compressed to 1, 1-1/2, and 2 inches with the loading required

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for these deflections noted. The results shall be compared to the requirements in 3.3.23. The manufacturer of the spring unit shall supply a certificate that the units supplied meet the requirements for regular, firm or extra firm, as required.

4.6 Flammability. Flammability tests shall be performed in accordance with all the provisions of the Consumer Product Safety Commission Flammability Standard for Mattresses FF-4-72 (16 CFR Part 1632). A contractor's certificate of compliance shall be furnished and will be accepted for the stated requirements. Certification shall include construction and material details used to meet the standard.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Mattresses shall be packaged in accordance with the applicable requirements of MIL-M-3996.

5.1.2 Civil agencies. For civil agency procurements, where level B packaging is specified in the contract or purchase order, the requirements in 5.1.1 (level A) shall apply.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Levels A, B, and C. Mattresses shall be packed in accordance with the applicable requirements of MIL-M-3996.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with FED-STD-123.

5.3.1.1 Special marking. The following marking shall also appear on at least two side edges of each shipping container in letters not less than 1-1/2 inches in height:

STOW FLAT - DO NOT STOW ON END OR SIDE

The shipping container shall have a color coded decal, stenciling, or other suitable means, affixed to the both end-edges of the container to identify the degree of firmness of the mattress. The marking shall be clearly visible from a distance of 10 feet.

5.3.2 Military requirements. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-M-3996.

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6. NOTES

6.1 Intended use. These mattresses are intended for use by civil and military agencies of the Government.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type, class, size, and firmness of mattress required (see 1.2).
c. Ticking style (see 3.3.6).
d. When a first article sample is required (see 3.1, 4.3, and 6.3).
e. Whether medical label is required (see 3.7.3).
f. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical, as long as it meets the requirements of this specification (see 3.3).

6.5 Flame retardant and smolder resistant finishes and materials. (see 3.3.5.3 and 3.5.7). Approval of such components and combinations is the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, and is based on extensive tests including those for toxicity performed by Office of the Surgeon General which are not set forth in this specification. Because of the time necessary to conduct full evaluation, only those chemical treatments and materials already approved and so listed in the invitation for bid or request for proposal shall be considered acceptable for the related procurement.

6.6 Bonding agents. (see 3.3.5.2). Approval of bonding agents for use in battings covered by this specification is the responsibility of the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014. Approval is based on more extensive evaluations, including those of toxicity, which are not set forth in the specification. Because of the time necessary to conduct full evaluation, only those bonding agents already approved and so listed in the invitation for bids or request for proposal shall be considered acceptable for the related procurement.

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6.6.1 Submittal for approval. Request for approval should be submitted to the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, attention of STRNC-ESE. It should be accompanied by approximately 5 square yards of a minimum of 2 oz/yd polyester batting manufactured with the proposed bonding agent together with applicable test data. Information is to be furnished regarding the commercial designation and chemical description of the bonding agent. (Approved lists will indicate only the commercial designation). Chemical identification with products already approved should be made, if applicable, and any available data on skin irritancy or toxicity should be submitted. If applicable, a statement should be included that as used the resin is well cured and that no materials with known high toxicity are used prior to cure.

6.7 Edge binding tape. (See 3.3.11.1). The polyester/polypropylene tape which met the requirements of 3.3.11.1 was based on 5/8 inch wide tape construction using 50 ends of 2/20 polyester and 24 picks of 360 denier polypropylene.

6.8 Proposed Voluntary Product Standard. The Proposed Voluntary Product Standard TS 222 for Bedding Products and Components (Mattresses, Foundations, Beds, Bed Frames, and Headboards) is being coordinated by the National Bureau of Standards for comments. (Application for copies of the proposed standard should be addressed to Standards Development Services, A165 Technology Building, National Bureau of Standards, Washington, DC 20234.)

6.9 Innerspring unit dimensions. The table below outlines innerspring unit dimensions which have been found suitable in fabricating mattresses to the finished sizes specified in tables III and IV.

TABLE XV. Innerspring unit dimensions

	Type II						Type III	
	Regular		Firm and Extra firm				All firmness	
Size No,	Length (inches)	Width (inches)	Length (inches)	Width (inches)	Length (inches)	Width (inches)	Length (inches)	Width (inches)
1	75	28	74	27-1/2 or	75	28	74	27-1/2
2	-	-	-	-	-	-	76	31-1/2
3A	-	-	-	-	-	-	73	33-1/2
4	74	35	73	34-1/2 or	74	35	73	34-1/2
5	77	35	76	34-1/2 or	77	35	76	34-1/2
6	83	35	83	35	83	35	82	34-1/2
6A	-	-	-	-	-	-	82	33-1/2
7	74	37	73	36-1/2 or	74	37	73	36-1/2
8	77	38	76	37-1/2 or	77	38	76	37-1/2
9	-	-	-	-	-	-	76	40
10	-	-	-	-	-	-	76	45-1/2

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TABLE XV. Innerspring unit dimensions (cont'd)

Size No.	Type II Regular Length (inches)	Type II Regular Width (inches)	Type II Firm and Extra firm Length (inches)	Type II Firm and Extra firm Width (inches)	Type III All firmness Length (inches)	Type III All firmness Width (inches)
11	-	-	-	-	71-1/2	51-1/2
12	74	52	73	51-1/2 or 74	52	73 51-1/2
13	79 or 79	33 or 35	78	34-1/2 or 79	35	78 34-1/2
13A	79	34-1/2	-	-	-	- 78 33-1/2
14	79	37	78	36-1/2 or 79	37	78 36-1/2
15	79	52	79	52	79	52 78 51-1/2
16	79	59	79	59	79	59 78 58-1/2

MILITARY INTERESTS:

Custodians

Army - GL
Navy - YD

Review Activities

Army - MD
Navy - MC
DLA - CT

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity

Army - GL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
V-M-96J	Mattresses, Bed, Innerspring

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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